

# Handling Ambiguous SSA Cases Completely

## Answer Key

Precalculus    Trigonometry and Vectors

### Quick Answers

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|--------------------------------|--------------------------------|
| 1. 38.68, 5.00                 | 6. 90, 6.00, 10.39             |
| 2. 58.99, 7.71, 1.29           | 7. 36.06, 50.46                |
| 3. 63.15, 116.85, 15.53, 7.40  | 8. 53.13, 126.87, 29.78, 11.78 |
| 4. 31.35, 196.65, 13.22        | 9. 64.56, 115.44, 27.29, 10.11 |
| 5. 46.58, 133.42, 52.82, 18.46 | 10. 61.10, 12.26               |
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### Curriculum

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**Level:** Precalculus    Trigonometry and Vectors

HSG-SRT.D.10, HSG-SRT.D.11 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

*Difficulty 1–5 of 5 across 33 tagged checks.*

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### Common wrong answers

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1. *If they got 4.00: computed a  $\sin A$  instead of  $b \sin A$ , testing the wrong side against the height*
  3. *If they got 63.15: stopped at the acute angle and never reported the second triangle*
  5. *If they got 46.58: reported only the acute angle and never checked whether the supplement also fits*
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**1. Case test:**  $A = 30^\circ$ ,  $b = 10$ ,  $a = 8$ .

**Step 1.** Part (a): the height is the perpendicular from  $C$  to the base line,  $h = b \sin A = 10 \sin 30^\circ$ .  $h = 5.00$

**Step 2.** Part (b):  $h = 5 < a = 8 < b = 10$ , so the swinging side is long enough to reach the base line but short enough to reach it twice — two triangles.

**Step 3.** The law of sines gives  $\sin B = \frac{b \sin A}{a} = \frac{5}{8} = 0.625$ , so  $B = \arcsin 0.625$ .

$B = 38.68^\circ$

*Why  $h$  comes first:* it is the only number that decides the case, and it costs one multiplication. Students who skip it end up discovering the second triangle by luck.

**2. Two sightings:**  $A = 40^\circ$ ,  $b = 12$ , first  $a = 9$ .

**Step 1.** Part (a):  $h = 12 \sin 40^\circ$ .

$$h = 7.71$$

**Step 2.** Since  $7.71 < 9 < 12$ , the measurement is ambiguous: two triangles. The acute solution comes from  $\sin B = \frac{12 \sin 40^\circ}{9} = 0.8571$ .

$$B = 58.99^\circ$$

**Step 3.** Part (b): with  $a = 6$  the same rule gives  $\sin B = \frac{12 \sin 40^\circ}{6}$ .

$$\sin B = 1.29$$

**Step 4.** No angle has a sine above 1, so no triangle exists — which matches the height test, since  $a = 6$  is shorter than  $h = 7.71$  and the swinging side cannot reach the base line at all. The second sighting must be a measurement error.

*Grading note:* accept either justification in part (b) — the impossible sine or the comparison with  $h$ . They are the same fact stated two ways.

**3. Both triangles:**  $A = 35^\circ$ ,  $b = 14$ ,  $a = 9$ .

**Step 1.**  $h = 14 \sin 35^\circ = 8.03$ , and  $8.03 < 9 < 14$ , so expect two triangles. From  $\sin B = \frac{14 \sin 35^\circ}{9} = 0.8922$ :

$$B_1 = 63.15^\circ$$

**Step 2.** The supplement has the same sine:  $B_2 = 180^\circ - 63.15^\circ$ . Test it:  $A + B_2 = 35^\circ + 116.85^\circ = 151.85^\circ < 180^\circ$ , so a third angle of  $28.15^\circ$  remains and the second triangle is real.

$$B_2 = 116.85^\circ$$

**Step 3.** Triangle 1:  $C_1 = 180^\circ - 35^\circ - 63.15^\circ = 81.85^\circ$ , and  $c_1 = \frac{a \sin C_1}{\sin A} = \frac{9 \sin 81.85^\circ}{\sin 35^\circ}$ .

$$c_1 = 15.53$$

**Step 4.** Triangle 2:  $C_2 = 180^\circ - 35^\circ - 116.85^\circ = 28.15^\circ$ , and  $c_2 = \frac{9 \sin 28.15^\circ}{\sin 35^\circ}$ .

$$c_2 = 7.40$$

*The check that catches everything:* in each triangle the largest side must face the largest angle. Triangle 1 has  $C_1$  largest and  $c_1 = 15.53$  largest; Triangle 2 has  $B_2$  largest and  $b = 14$  largest. Both pass.

**4. Only one:**  $A = 48^\circ$ ,  $b = 7$ ,  $a = 10$ .

**Step 1.** Part (a):  $\sin B = \frac{7 \sin 48^\circ}{10} = 0.5202$ .

$$B = 31.35^\circ$$

**Step 2.** Part (b): the obtuse candidate is  $180^\circ - 31.35^\circ = 148.65^\circ$ , and adding the given angle gives  $48^\circ + 148.65^\circ$ .

$$A + B_2 = 196.65^\circ$$

**Step 3.** That exceeds  $180^\circ$ , so no triangle can contain both angles and the obtuse candidate is rejected. This is what  $a \geq b$  always forces: the side opposite the larger angle is longer, so  $a > b$  makes  $A > B$  and  $B$  cannot be obtuse.

**Step 4.** Part (c):  $C = 180^\circ - 48^\circ - 31.35^\circ = 100.65^\circ$  and  $c = \frac{10 \sin 100.65^\circ}{\sin 48^\circ}$ .

$$c = 13.22$$

*Worth saying aloud:* “one triangle” here is not luck — it is guaranteed the moment  $a \geq b$ , before any arithmetic.

**5. Search and rescue:**  $A = 27^\circ$ ,  $b = 40$ ,  $a = 25$ .

**Step 1.**  $h = 40 \sin 27^\circ = 18.16$ , and  $18.16 < 25 < 40$ : the bearings allow two positions. From  $\sin B = \frac{40 \sin 27^\circ}{25} = 0.7264$ :

$$B_1 = 46.58^\circ$$

**Step 2.** The supplement passes the test ( $27^\circ + 133.42^\circ = 160.42^\circ < 180^\circ$ ), so it is a genuine second position.

$$B_2 = 133.42^\circ$$

**Step 3.** Far position:  $C_1 = 180^\circ - 27^\circ - 46.58^\circ = 106.42^\circ$ , so  $c_1 = \frac{25 \sin 106.42^\circ}{\sin 27^\circ}$ .

$$c_1 = 52.82 \text{ km}$$

**Step 4.** Near position:  $C_2 = 180^\circ - 27^\circ - 133.42^\circ = 19.58^\circ$ , so  $c_2 = \frac{25 \sin 19.58^\circ}{\sin 27^\circ}$ .

$$c_2 = 18.46 \text{ km}$$

*Why this matters operationally:* the two answers are 52.82 km and 18.46 km along the same shoreline. Reporting only the first is not a rounding error — it is a search in the wrong place, and it is exactly what the law of sines will hand you if you never test the supplement.

**6. The borderline:**  $A = 30^\circ$ ,  $b = 12$ ,  $a = 6$ .

**Step 1.** Part (a):  $h = 12 \sin 30^\circ$ .

$$h = 6.00$$

**Step 2.** Here  $a = h$  exactly. The swinging side just reaches the base line, meeting it at the foot of the perpendicular, so the angle at  $B$  is a right angle:  $\sin B = \frac{12 \sin 30^\circ}{6} = 1$ .

$$B = 90^\circ$$

**Step 3.** Part (b): the supplement of  $90^\circ$  is  $90^\circ$  again — there is no second angle to find. The two solutions of the ambiguous case have merged into one.

**Step 4.** Part (c):  $C = 180^\circ - 30^\circ - 90^\circ = 60^\circ$ , so  $c = \frac{6 \sin 60^\circ}{\sin 30^\circ}$ .

$$c = 10.39$$

*The boundary case is the useful one:* it is the exact moment the two triangles of the ambiguous case collapse into each other, which is why  $a = h$  separates “none” from “two”.

**7. Surveying:**  $A = 62^\circ$ ,  $b = 30$ ,  $a = 45$ .

**Step 1.** Part (a):  $a = 45$  is longer than  $b = 30$ , so the angle opposite  $a$  is the larger of the two and  $B$  must be acute. No supplement test is needed — one triangle, guaranteed. Then  $\sin B = \frac{30 \sin 62^\circ}{45} = 0.5886$ .

$$B = 36.06^\circ$$

**Step 2.** Part (b):  $C = 180^\circ - 62^\circ - 36.06^\circ = 81.94^\circ$ , so  $c = \frac{45 \sin 81.94^\circ}{\sin 62^\circ}$ .

$$c = 50.46 \text{ m}$$

*Check:* the largest angle here is  $C = 81.94^\circ$  and the largest side is  $c = 50.46$  m. Consistent.

**8. Both triangles again:**  $A = 30^\circ$ ,  $b = 24$ ,  $a = 15$ .

**Step 1.**  $h = 24 \sin 30^\circ = 12.00$ , and  $12 < 15 < 24$ : two triangles. From  $\sin B = \frac{24 \sin 30^\circ}{15} = 0.8000$ :

$$B_1 = 53.13^\circ$$

**Step 2.** The supplement passes ( $30^\circ + 126.87^\circ = 156.87^\circ$ ).

$$B_2 = 126.87^\circ$$

**Step 3.** Triangle 1:  $C_1 = 96.87^\circ$  and  $c_1 = \frac{15 \sin 96.87^\circ}{\sin 30^\circ}$ .

$$c_1 = 29.78$$

**Step 4.** Triangle 2:  $C_2 = 23.13^\circ$  and  $c_2 = \frac{15 \sin 23.13^\circ}{\sin 30^\circ}$ .

$$c_2 = 11.78$$

**Step 5.** Part (c): in the figure the solid apex is the far one, giving the long base  $c_1$ ; the dashed apex sits nearer  $A$ , giving the short base  $c_2$ . A small given angle  $A$  stretches the difference, which is why 29.78 and 11.78 are so far apart.

*Fast sanity check:* the two bases differ by  $2b \cos B_1$  — geometry worth showing students who suspect one of the answers is a slip.

**9. Two possible positions:**  $A = 44^\circ$ ,  $b = 26$ ,  $a = 20$ .

**Step 1.**  $h = 26 \sin 44^\circ = 18.06 < 20 < 26$ , so both positions are geometrically valid. From  $\sin B = \frac{26 \sin 44^\circ}{20} = 0.9031$ :

$$B_1 = 64.56^\circ$$

**Step 2.** The supplement passes ( $44^\circ + 115.44^\circ = 159.44^\circ$ ).

$$B_2 = 115.44^\circ$$

**Step 3.** With  $C_1 = 71.44^\circ$ :  $c_1 = \frac{20 \sin 71.44^\circ}{\sin 44^\circ}$ .

$$c_1 = 27.29$$

**Step 4.** With  $C_2 = 20.56^\circ$ :  $c_2 = \frac{20 \sin 20.56^\circ}{\sin 44^\circ}$ .

$$c_2 = 10.11$$

**Step 5.** Part (b) — open response, read the reasoning. The obtuse solution  $B_2 = 115.44^\circ$  puts the marker between the control point and the drone (the near apex,  $c_2 = 10.11$ ), so “the drone is beyond the marker” rules it out and leaves the acute triangle with  $c_1 = 27.29$ .

*What a full answer names:* any single extra measurement that separates the two — the distance  $c$  itself, the angle at  $C$ , or whether the angle at  $B$  is acute or obtuse. Accept any of these; the point is that ONE more observation is enough, and that without it both answers must be reported.

**10. Challenge — the whole case rule:**  $A = 50^\circ$ ,  $b = 16$ .

**Step 1.** Part (a): with  $a = 14$ ,  $\sin B = \frac{16 \sin 50^\circ}{14} = 0.8755$ .

$$B_1 = 61.10^\circ$$

**Step 2.** The supplement  $118.90^\circ$  also passes ( $50^\circ + 118.90^\circ = 168.90^\circ < 180^\circ$ ), so  $a = 14$  gives two triangles.

**Step 3.** Part (b):  $h = 16 \sin 50^\circ$ .

$$h = 12.26$$

**Step 4.** Part (c) — open response, read the reasoning. The complete rule for  $A = 50^\circ$ ,  $b = 16$ :

- $a < 12.26$ : no triangle. The swinging side is shorter than the perpendicular and never reaches the base line.
- $a = 12.26$ : exactly one, the right triangle where the side meets the base line at the foot of the perpendicular.
- $12.26 < a < 16$ : two triangles. The side is long enough to cross the base line on both sides of the perpendicular.
- $a \geq 16$ : exactly one. The side now reaches past the vertex  $A$ , so only the far intersection lies on the correct side of  $A$ ; equivalently  $a \geq b$  forces  $B$  acute.

*Accept* any statement of these four bands with the two boundary values 12.26 and 16 identified, justified either by the swinging side or by the sine test ( $\sin B > 1$ ,  $= 1$ , or two admissible angles).